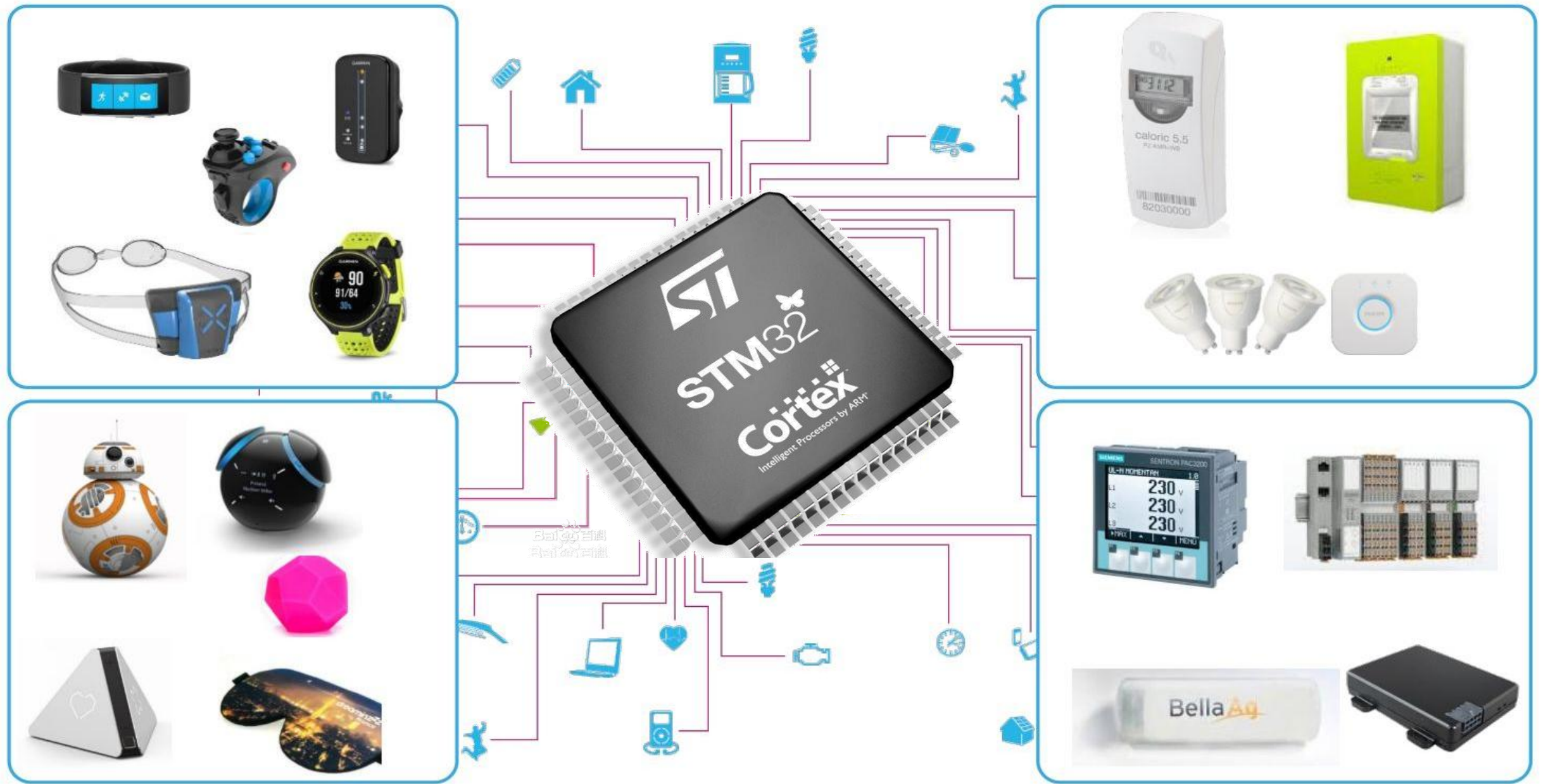
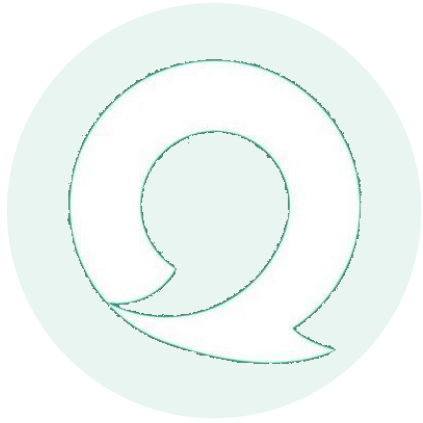




Instructor : WangXiaojing

Email : wangxj@sustech.edu.cn



Lecture1

Introduction and Preparation

Topics

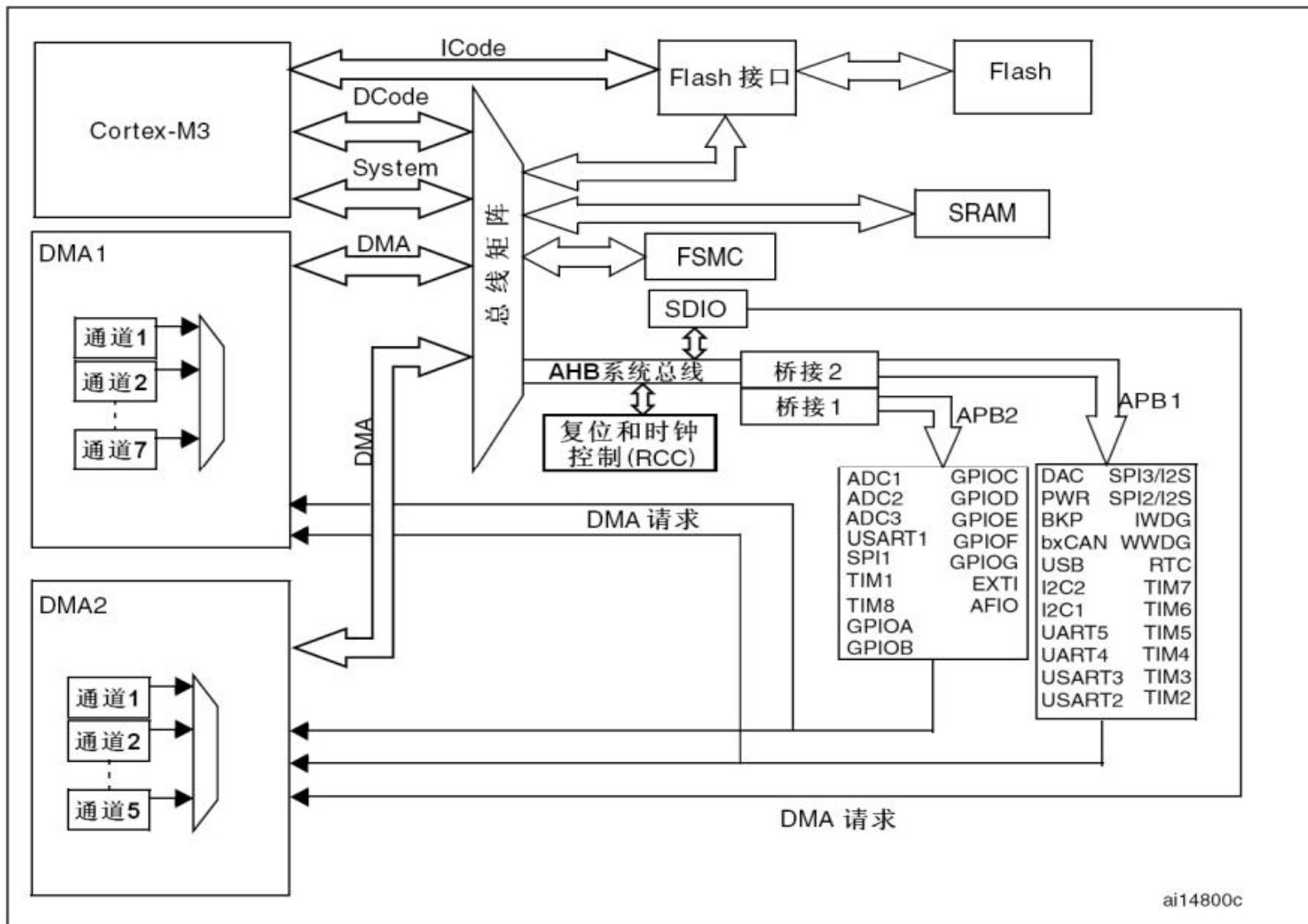
- Overview and Software
- Some Important learning materials
- STM32's "Hello World"--Light a LED
- Programming mode of STM32 & Code Analysis

Part I : Overview and Software

What's & Why STM32



Literally, ST is STMicroelectronics, M is the abbreviation of Microcontrollers, and 32 represents 32 bits. So the integrated understanding is: STM32 refers to the 32-bit microcontroller developed by ST company. In today's 32-bit controller, STM32 can be said to be the most shining star, which is favored by most engineers and the market.



STM32F1系列系统架构图

Cortex-M3 (IP) : 是一款ARM处理器内核 ;

ICode : 指令总线

Dcode : 数据总线

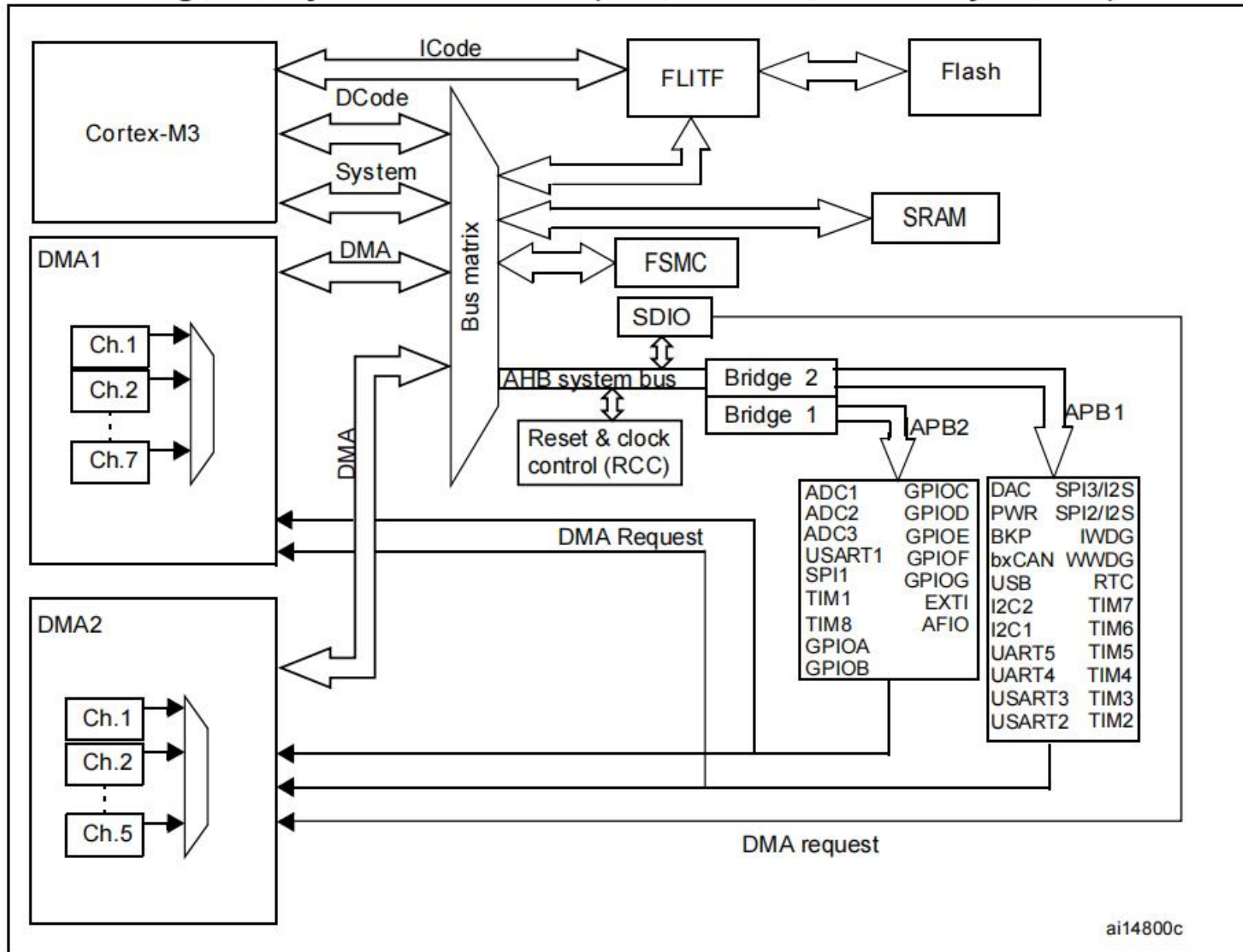
System : 系统总线

STM32 (SoC) : 是一款ARM内核的处理器芯片。

权威指南P18

<https://blog.csdn.net/ybhuangfugui/article/details/104471010>

Figure 1. System architecture (low-, medium-, XL-density devices)



Cortex-M3 (IP) : the central processing unit (CPU) ;

ICode : Instructions Bus

Dcode : Data Bus

System : System Bus

STM32 (SoC) : a microcontroller chip whose central processing unit is Cortex-M3

The Cortex-M3 Processor vs Cortex-M3-Based MCUs

The Cortex-M3 processor is the central processing unit (CPU) of a microcontroller chip. In addition, a number of other components are required for the whole Cortex-M3 processor-based microcontroller. After chip manufacturers license the Cortex-M3 processor, they can put the Cortex-M3 processor in their silicon designs, adding memory, peripherals, input/output (I/O), and other features. Cortex-M3 processor-based chips from different manufacturers will have different memory sizes, types, peripherals, and features. This book focuses on the architecture of the processor core. For details about the rest of the chip, please check the particular chip manufacturer's documentation.

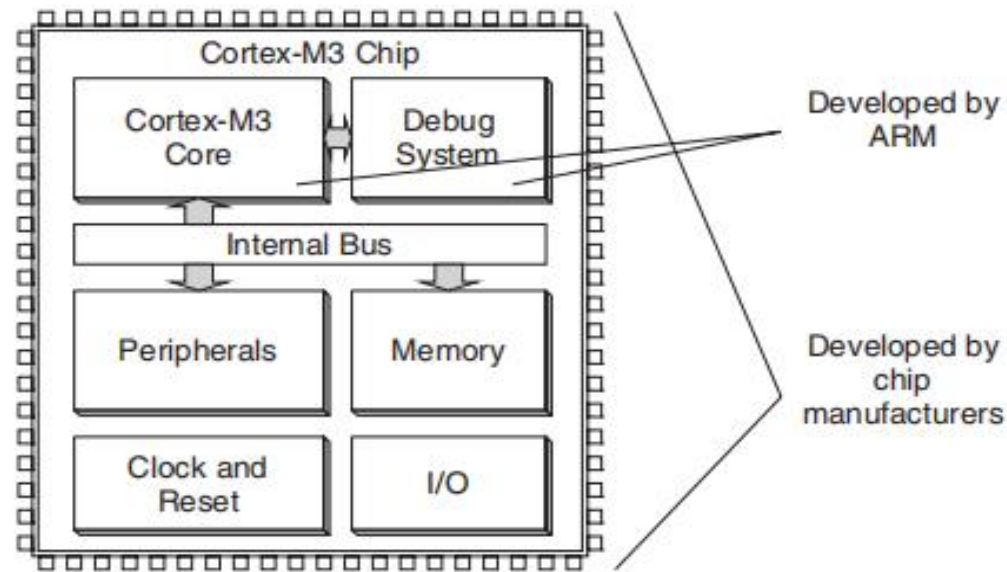


Figure 1.1 The Cortex-M3 Processor vs the Cortex-M3-Based MCU

How to use STM32



Communicate with it

Programming in a
language it can Understand

C

Corresponding compilation
environment

Keil5 (MDK)

C Language



Visual Studio

C language is just a tool, so we don't study it systematically. We just learn the relevant knowledge where we use it, for the purpose of understanding the code.

Learn by yourself!

An Overview of C Language

The basic knowledge points of C language introduction are as follows:

- 1.The naming of C language;
- 2.Variable and assignment;
- 3.Basic data type;
- 4.Format output statements;
- 5.Compulsory Type Conversion(Cast type);
- 6.Constant;
- 7Automatic type conversion.
- 8.Constructs

KEIL5(MDK)

www.keil.com/dd2/pack

> STMicroelectronics BlueNRG-2 Series Device Support	DFP	1.0.1	Download	
> STMicroelectronics BlueNRG-LP Series Device Support	DFP	2.0.0	Download	
> STMicroelectronics Nucleo Boards Support and Examples	BSP	1.8.0	Download	
> STMicroelectronics STM32F0 Series Device Support, Drivers and	BSP	DFP	2.1.1	Download
> STMicroelectronics STM32F1 Series Device Support, Drivers and	BSP	DFP	2.4.0	Download
> STMicroelectronics STM32F2 Series Device Support, Drivers and	BSP	DFP	2.10.0	Download
> STMicroelectronics STM32F3 Series Device Support and Examples	BSP	DFP	2.2.2	Download
> STMicroelectronics STM32F4 Series Device Support, Drivers and	BSP	DFP	2.16.0	Download
> STMicroelectronics STM32F7 Series Device Support, Drivers and	BSP	DFP	2.14.1	Download

Basic package : Necessary
Chip package : Optional
(According to your board
STM32F103)

Precautions

MDK(keil5) installation precautions :

- 1:The installation path cannot be in Chinese, but must be in English;
- 2:The installation directory cannot conflict with keil or keil4 of 51, and its directory must be separated;
- 3:The installation of keil5 is one more step than that of keil4. The chip-package must be installed, otherwise it cannot be used;
- 4:If the reareinexplicable errors when using, first lookup the information on the Internet, don't panic;

By default, the software is a trial version , and can only compile code no more than 32K!! More than 32K needs to be registered!

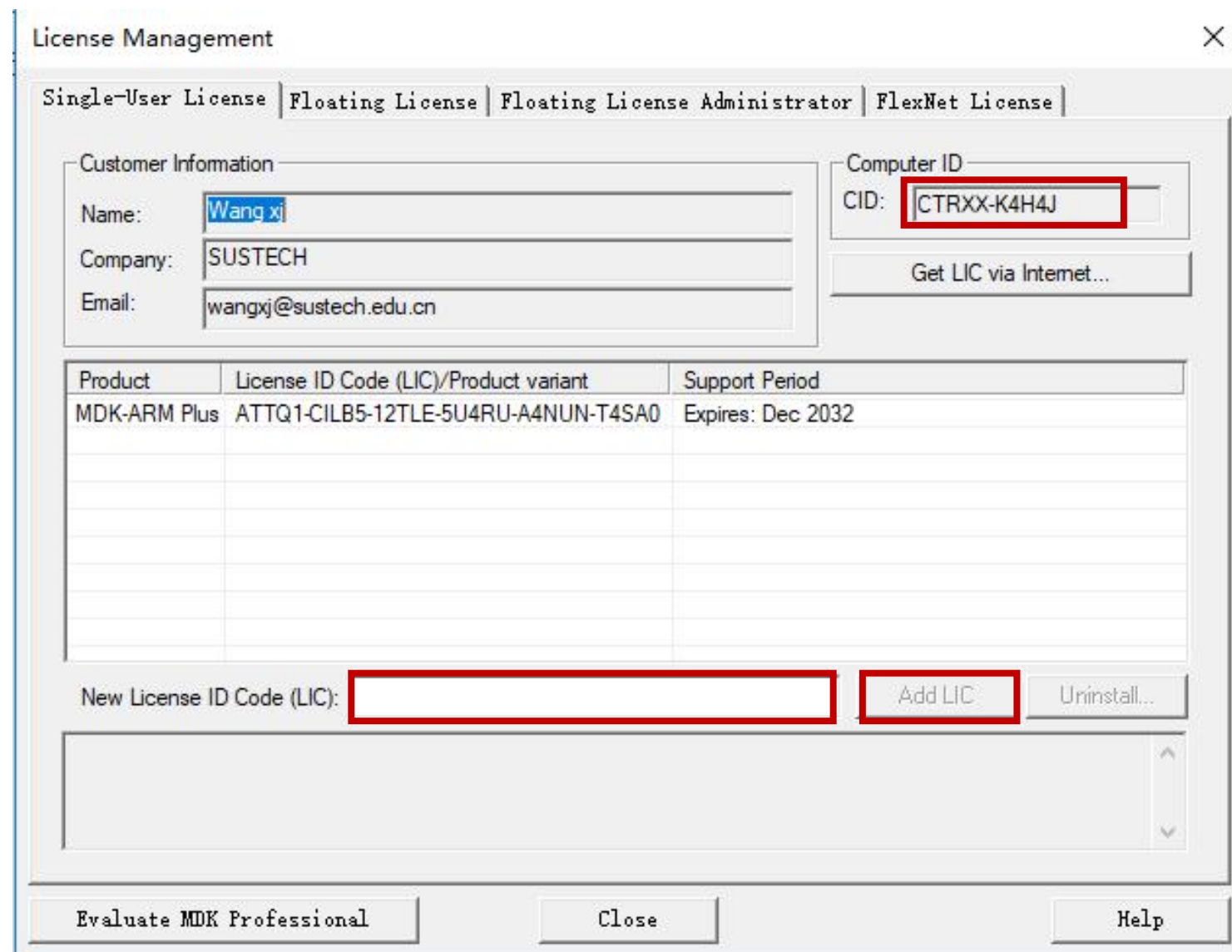
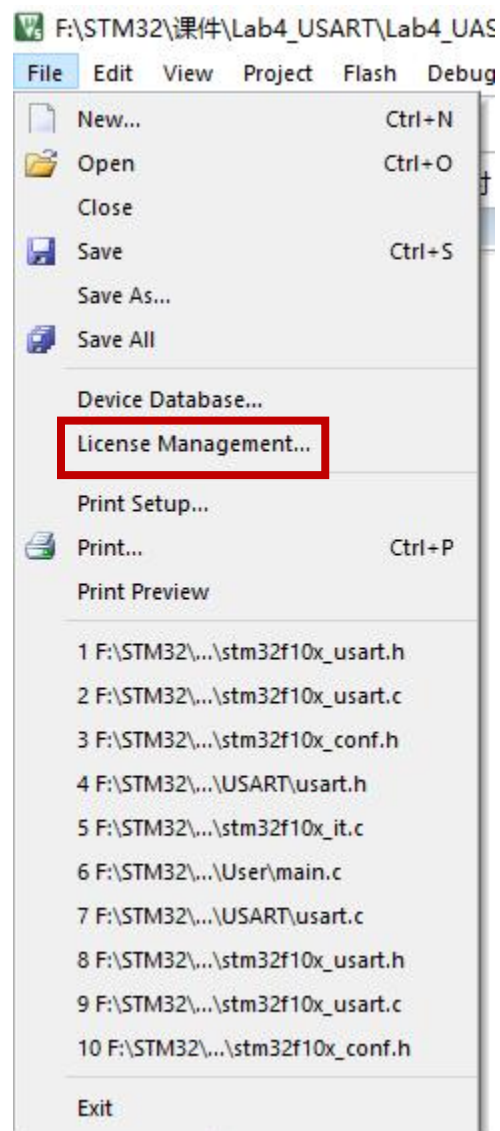
For students who can't afford to buy genuine ones, we need to crack them. There are many installation tutorials of keil5 on the Internet.

2	 Keil.STM32F1xx_DFP.2.4.0.pack	2022/1/18 15:59	uVision Software...	49,063 KB
3	 keygen (2).exe	2022/2/9 14:04	应用程序	498 KB
1	 MDK536.EXE	2022/1/13 17:38	应用程序	912,497 KB

1：按照顺序依次双击安装即可；

2：基本都是默认设置，选择好路径即可----英文路径.后面芯片包安装的提示直接取消掉；

3：第三步中破解操作：以管理员身份运行KEIL5, file-->license management。keygen注册机，CID填入kei5 软件中license 管理页面的 CID，target 选择ARM，点击generate。把生成的一串数值粘贴到kei5 软件中license 管理页面的Add New License ID Code处，点击Add LIC。



License Management



Single-User License | Floating License | Floating License Administrator | FlexNet License

Customer Information

Name: Wang xi

Company: SUSTECH

Email: wangxj@sustech.edu.cn

Computer ID

CID: CTRXX-K4H4J

Get LIC via Internet...

Product	License ID Code (LIC)/Product variant	Support Period
MDK-ARM Plus	ATTQ1-CILB5-12TLE-5U4RU-A4NUN-T4SA0	Expires: Dec 2032

New License ID Code (LIC):

Add LIC

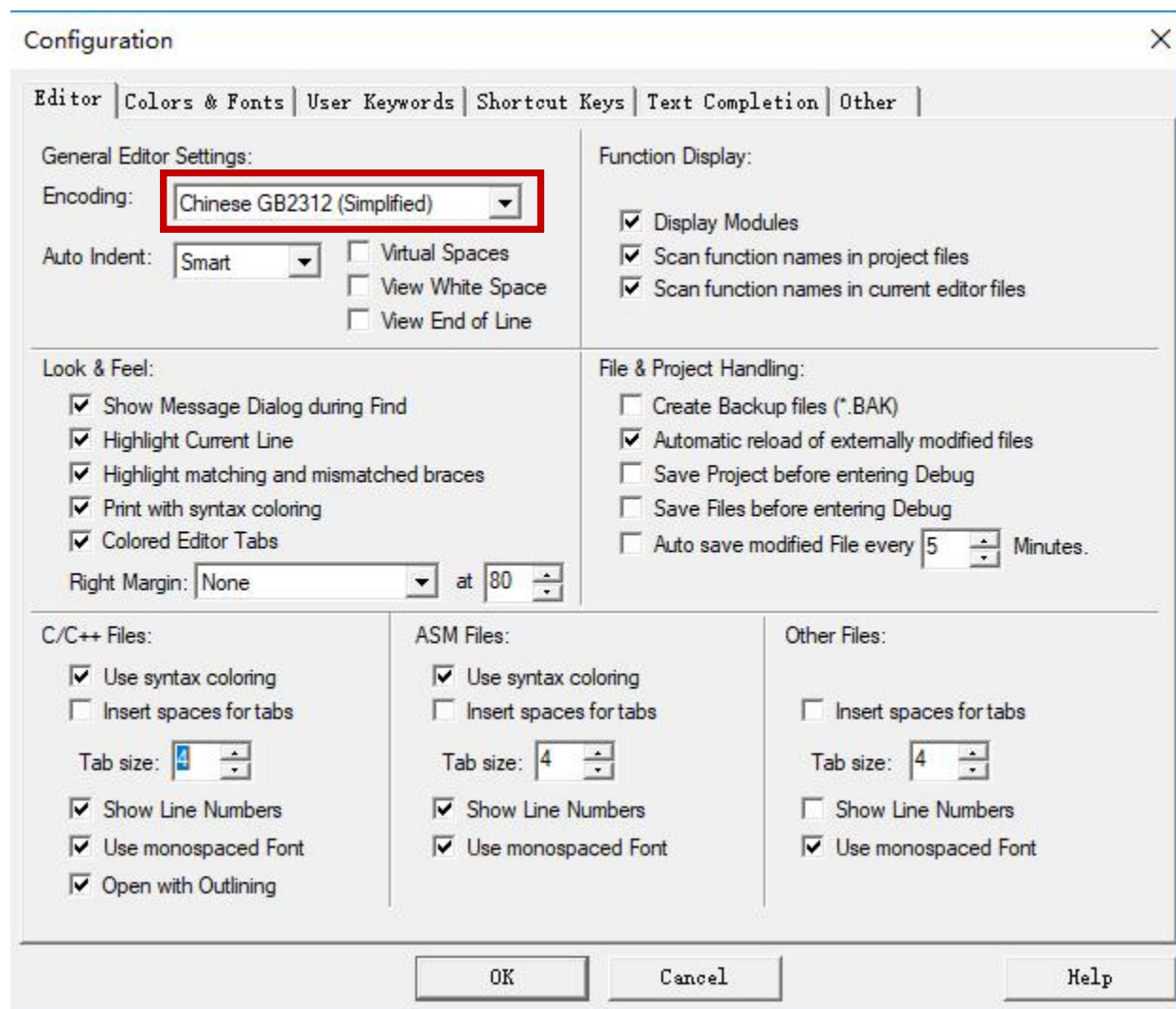
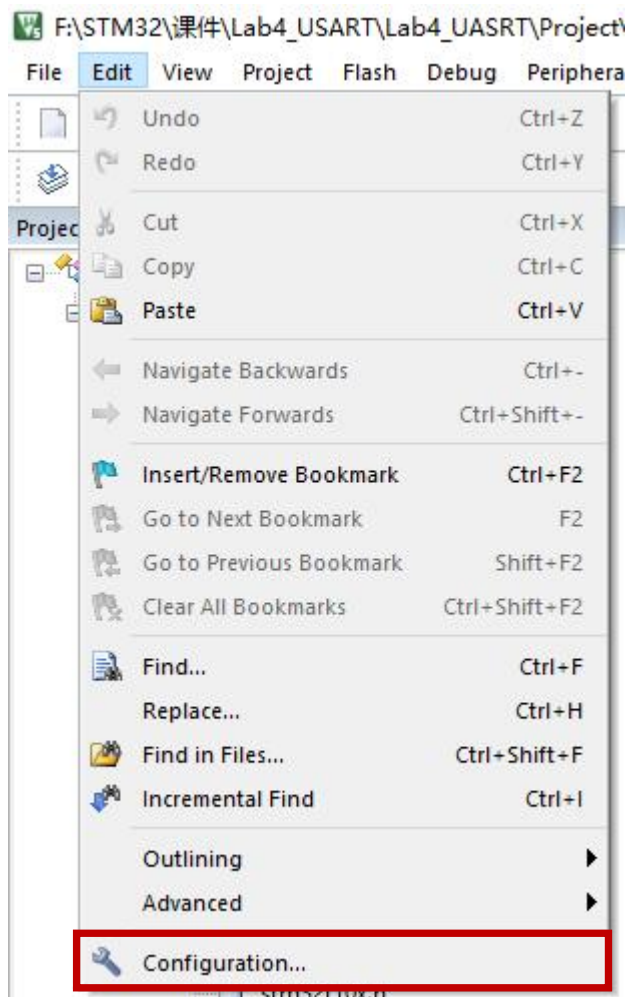
Uninstall...

Evaluate MDK Professional

Close

Help

Configuration of Keil5



Part II : Some Important learning materials

Datasheet : Device overview、Pin Discriptions、 Memory mapping、Electrical characteristics

Reference manual : How to configure registers

The Definitive Guide :Cortex-M3 Basics , The NVIC and Interrupt Control

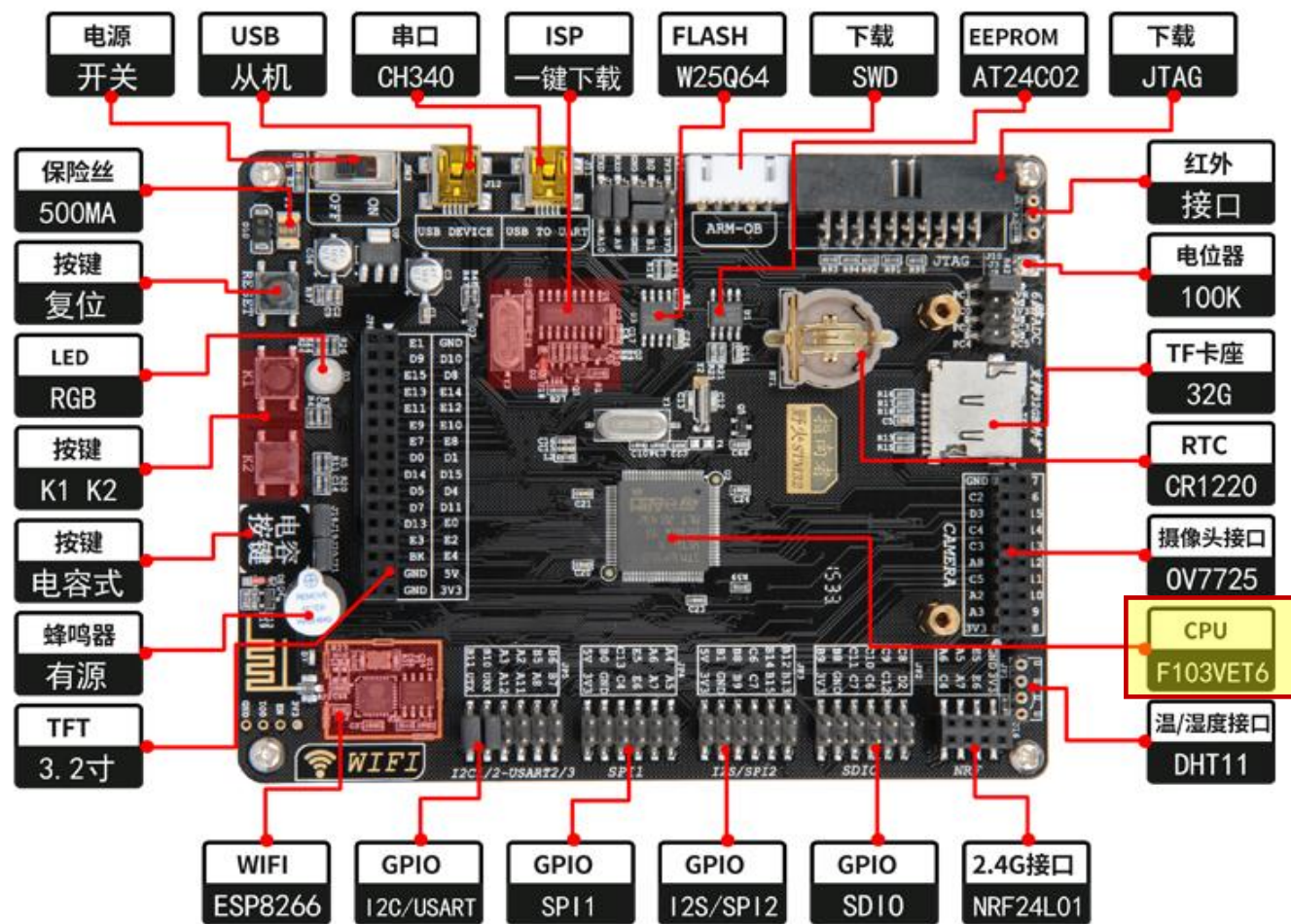
Schematic diagram : Hardware design

- ARM Cotrex-M3权威指南(英文).pdf
- Cortex-M3权威指南(中文).pdf
- stm32f103ve数据手册.pdf
- stm32f103-参考手册 (英文2022).pdf
- stm32f103-参考手册.pdf
- STM32f103参考手册_V10(中文).pdf

The screenshot shows the STMicroelectronics website's design resource page. The header includes the ST logo, navigation links like '关于STM32', '产品', '垂直应用', '设计资源', '活动与培训', 'ST中国大学计划', and '生态系统'. A search bar is on the right. The main content area is titled '设计资源 > 芯片文档'. It shows filter options: '类型/芯片文档/全部' and '产品/STM32 MCU/全部'. Under '类型', there are links for '芯片文档', '固件和软件', '评估开发板', '硬件开发工具', '中文译文', '实战经验', and '培训课件'. Under '产品', there are links for 'STM32 MCU', 'STM32 MPU', and 'STM8 MCU'. A list of specific documents is shown at the bottom, including '全部', 'STM32F0', 'STM32G0', 'STM32F1', 'STM32F2', 'STM32F3', 'STM32G4', 'STM32F4', 'STM32F7', 'STM32H7', 'STM32L0', 'STM32L1', 'STM32L4', 'STM32L4+', 'STM32L5', 'STM32U5', 'STM32WL', and 'STM32WB'.

<https://stm32.com.cn/Designresource/list/STM32%20MCU/document>

F103 指南者 | 硬件配置图



STM32系列产品命名规则

示例: STM32 F 103 C 8 T 6 A xxx

产品系列

STM32 = 基于ARM®的32位微控制器

产品类型

F = 通用类型

产品子系列

101 = 基本型

102 = USB基本型, USB 2.0全速设备

103 = 增强型

105或107 = 互联型

引脚数目

T = 36脚

C = 48脚

R = 64脚

V = 100脚

Z = 144脚

闪存存储器容量

4 = 16K字节的闪存存储器

6 = 32K字节的闪存存储器

8 = 64K字节的闪存存储器

B = 128K字节的闪存存储器

C = 256K字节的闪存存储器

D = 384K字节的闪存存储器

E = 512K字节的闪存存储器

封装

H = BGA

T = LQFP

U = VFQFPN

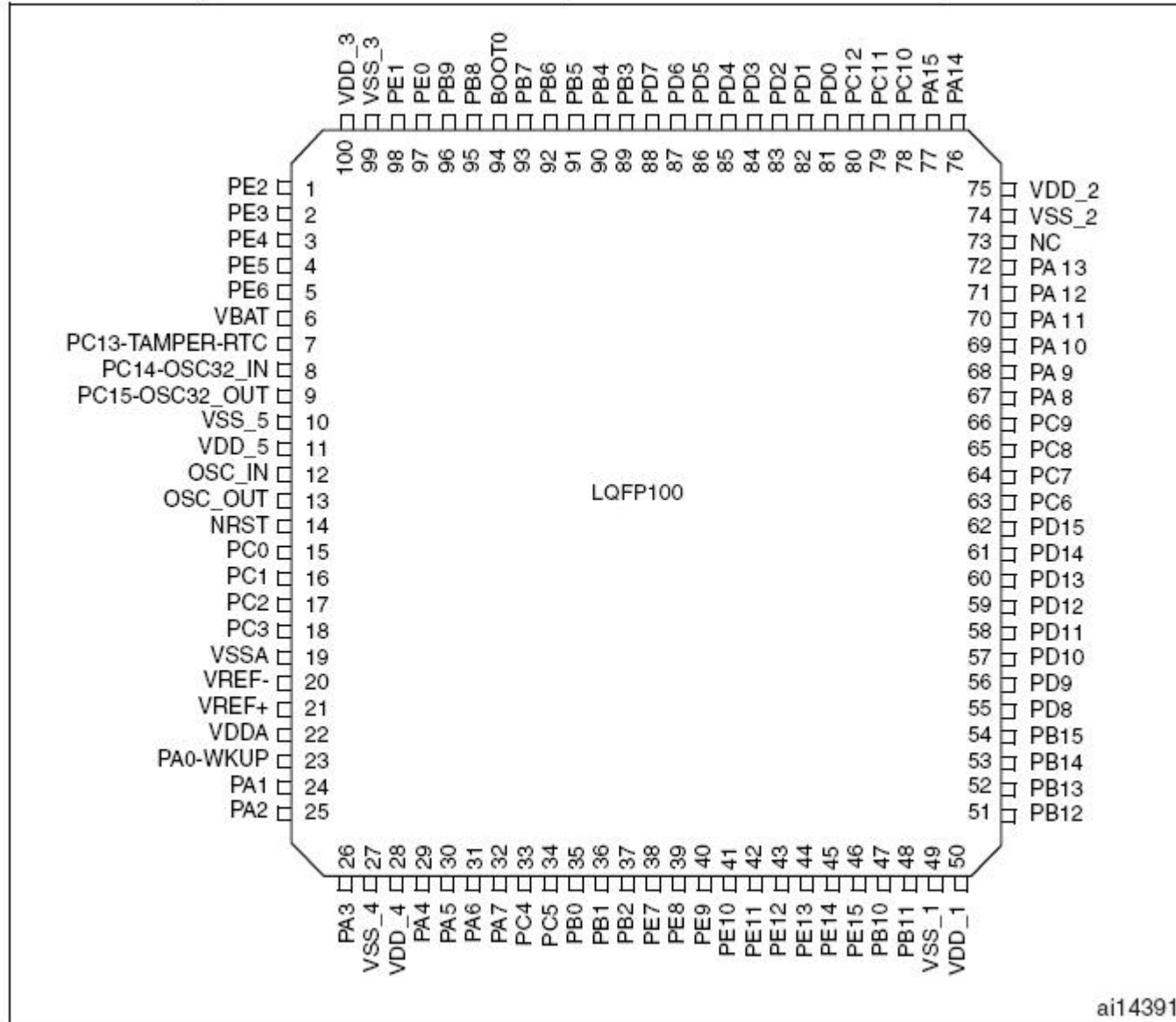
Y = WLCSP64

温度范围

6 = 工业级温度范围, -40°C~85°C



Figure 6. STM32F103xC/D/E performance line LQFP100 pinout



1. The above figure shows the package top view.

Datasheet

PA

PA0、 PA1...PA15

PB

PB0、 PB1...PB15

PC

PC0、 PC1...PC15

.

.

.

PE

PE0、 PE1...PE15

Table 5. High-density STM32F103xC/D/E pin definitions

Pins										Alternate functions ⁽⁴⁾	
LFBGA144	LFBGA100	WLCSP64	LQFP64	LQFP100	LQFP144	Pin name	Type ⁽¹⁾	I / O Level ⁽²⁾	Main function ⁽³⁾ (after reset)	Default	Remap
A3	A3	-	-	1	1	PE2	I/O	FT	PE2	TRACECK/ FSMC_A23	-
A2	B3	-	-	2	2	PE3	I/O	FT	PE3	TRACED0/FSMC_A19	-
B2	C3	-	-	3	3	PE4	I/O	FT	PE4	TRACED1/FSMC_A20	-
B3	D3	-	-	4	4	PE5	I/O	FT	PE5	TRACED2/FSMC_A21	-
B4	E3	-	-	5	5	PE6	I/O	FT	PE6	TRACED3/FSMC_A22	-
C2	B2	C6	1	6	6	V _{BAT}	S	-	V _{BAT}	-	-
A1	A2	C8	2	7	7	PC13-TAMPER-RTC ⁽⁵⁾	I/O	-	PC13 ⁽⁶⁾	TAMPER-RTC	-
M7	H5	-	-	38	58	PE7	I/O	FT	PE7	FSMC_D4	TIM1_ETR
L7	J5	-	-	39	59	PE8	I/O	FT	PE8	FSMC_D5	TIM1_CH1N
K7	K5	-	-	40	60	PE9	I/O	FT	PE9	FSMC_D6	TIM1_CH1

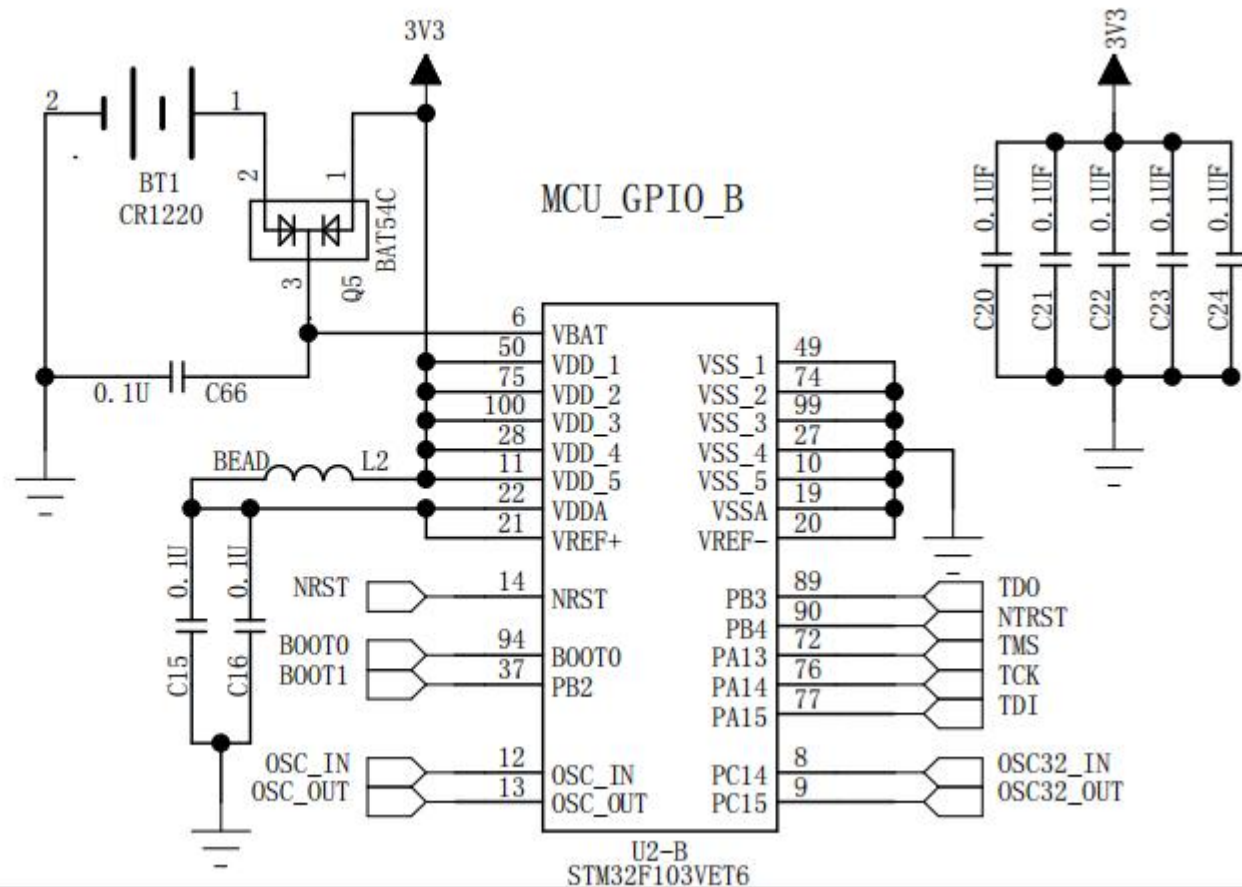
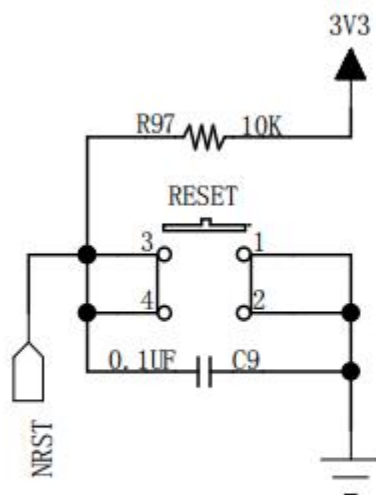
1. I = input, O = output, S = supply.

2. FT = 5 V tolerant.

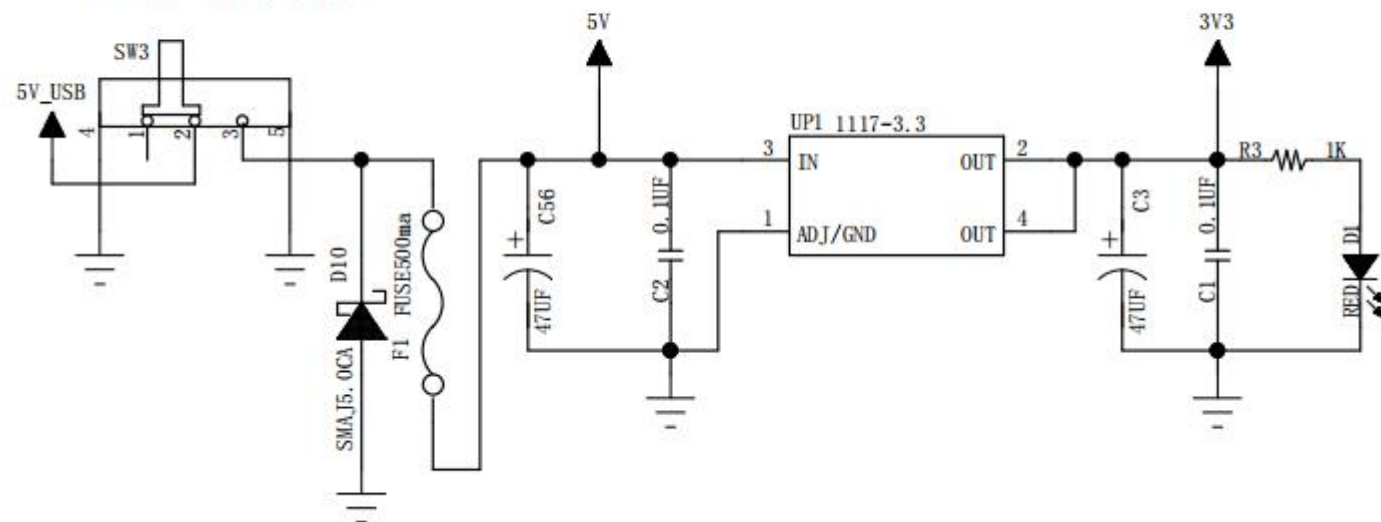
3. Function availability depends on the chosen device.

4. If several peripherals share the same I/O pin, to avoid conflict between these alternate functions only one peripheral should be enabled at a time through the peripheral clock enable bit (in the corresponding RCC peripheral clock enable register).

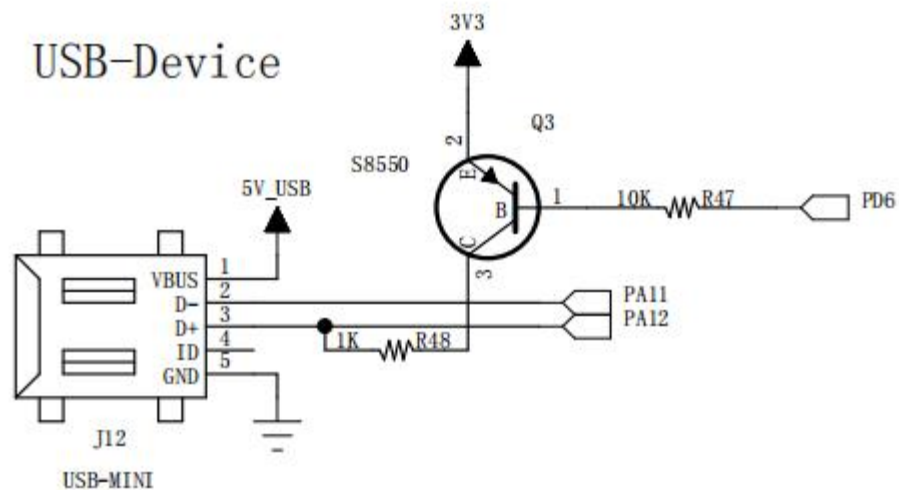
复位电路



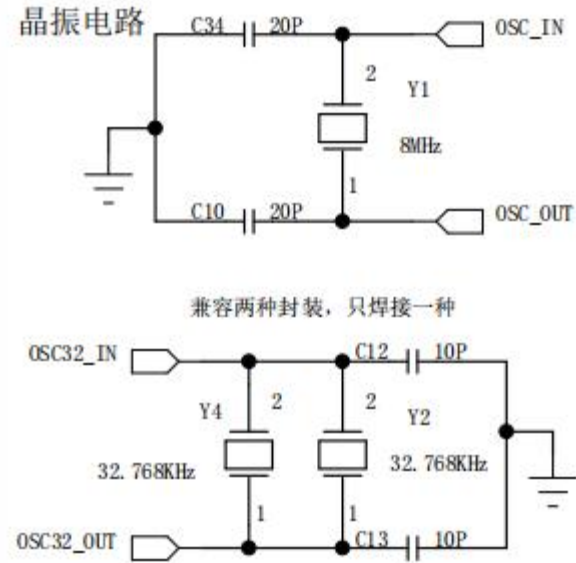
电源电路 USB-5V供电



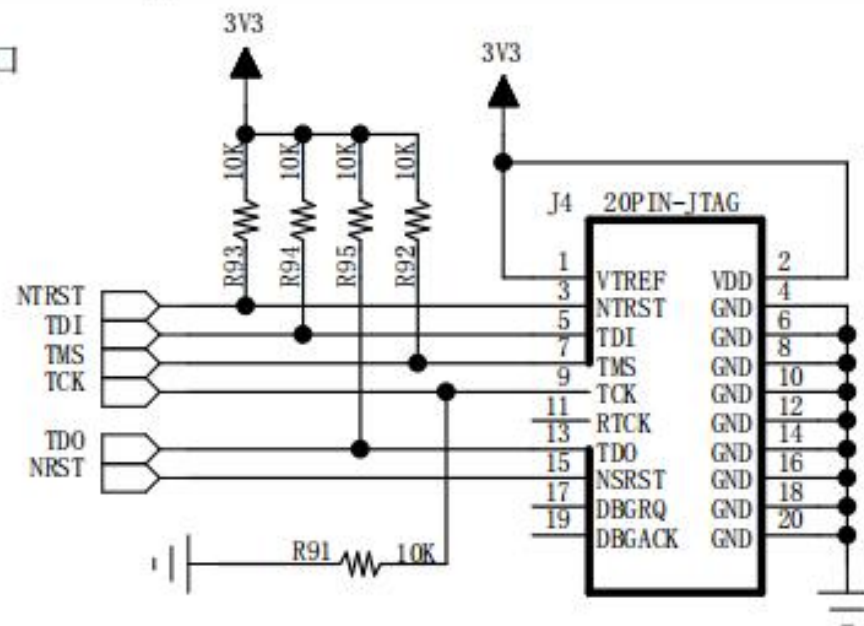
USB-Device



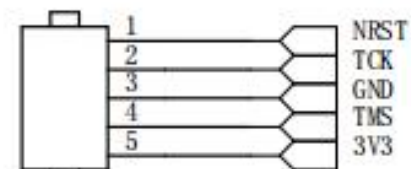
晶振电路



JTAG下载接口



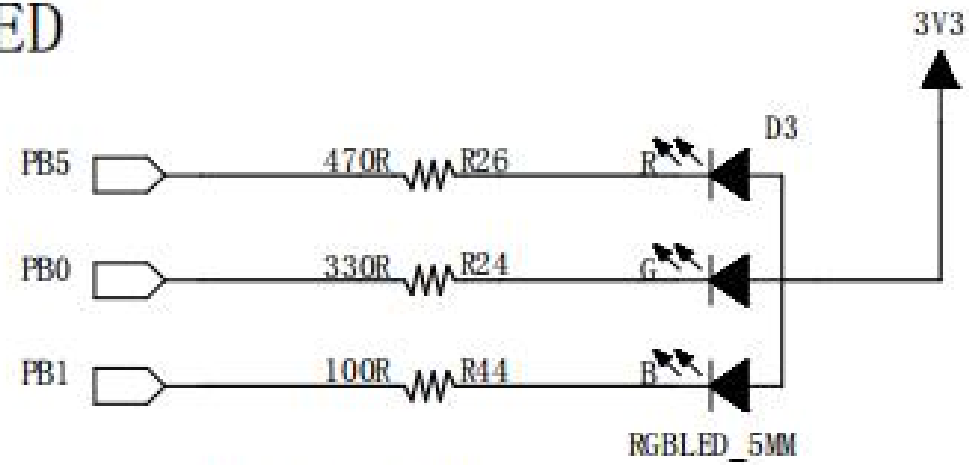
NRST 接 MCU 复位引脚 可以增加稳定性



J1
XH2.54-5P

3V3可对目标板供电

LED



TIM3_CH3	LED_G	PB0
TIM3_CH4	LED_B	PB1
TIM3_CH2	LED_R	PB5

Part III : STM32's "Hello World"--Light a LED

Demonstration

DebugConfig	2022/3/23 13:22	文件夹	
Listings	2022/3/23 13:22	文件夹	
Objects	2022/3/23 13:30	文件夹	
lab1.uvguix.lenovo	2022/3/23 13:30	LENOVO 文件	90 KB
lab1.uvoptx	2022/2/9 15:04	UVOPTX 文件	7 KB
lab1.uvprojx	2022/2/9 15:04	vision5 Project	14 KB
main.c	2022/3/23 13:29	C Source	1 KB
startup_stm32f10x_hd.s	2021/10/8 22:23	Assembler Source	15 KB

Lab1_reg1

LED_FWLib

..\Lab1_reg1

DebugConfig	2022/3/23 13:26	文件夹	
Listings	2022/3/23 13:26	文件夹	
Objects	2022/3/23 13:26	文件夹	
Template.uvguix.lenovo	2022/3/23 13:26	LENOVO 文件	90 KB
Template.uvoptx	2022/2/14 17:43	UVOPTX 文件	19 KB
Template.uvprojx	2022/2/14 17:43	vision5 Project	21 KB

..\LED_FWLib\Project

Download the program



ST LINK驱动及教程

amd64	2022/3/23 15:56	文件夹
x86	2022/3/23 15:56	文件夹
dpinst_amd64.exe	2010/2/9 4:36	应用程序
dpinst_x86.exe	2010/2/9 3:59	应用程序
stlink_dbg_winusb.inf	2014/1/21 17:03	安装信息
stlink_VCP.inf	2013/12/10 21:08	安装信息
stlink_winusb_install.bat	2013/5/15 22:33	Windows 批处理...
stlinkdbgwinusb_x64.cat	2014/1/21 17:14	安全目录
stlinkdbgwinusb_x86.cat	2014/1/21 17:14	安全目录
stlinkvcp_x64.cat	2013/12/10 21:08	安全目录
stlinkvcp_x86.cat	2013/12/10 21:09	安全目录

Options for Target 'Target 1'

Device | Target | Output | Listing | User | C/C++ (AC6) | Asm | Linker | Debug | Utilities

☐ Use Simulator [with restrictions](#) Settings☐ Limit Speed to Real-Time☒ Load Application at Startup ☒ Run to main()

Initialization File:

 ... Edit...

Restore Debug Session Settings

☒ Breakpoints ☒ Toolbox
☒ Watch Windows & Performance Analyzer
☒ Memory Display ☒ System Viewer

CPU DLL: Parameter:

SARMCM3.DLL -REMAP

Dialog DLL: Parameter:

DCM.DLL -pCM3

☐ Warn if outdated Executable is loaded☒ Use: ST-Link Debugger Settings☒ Load Application at Startup ☒ Run to main()

Initialization File:

 ... Edit...

Restore Debug Session Settings

☒ Breakpoints ☒ Toolbox
☒ Watch Windows ☒ Tracepoints
☒ Memory Display ☒ System Viewer

Driver DLL: Parameter:

SARMCM3.DLL

Dialog DLL: Parameter:

TCM.DLL -pCM3

☐ Warn if outdated Executable is loaded

Manage Component Viewer Description Files ...

OK

Cancel

Defaults

Help

Cortex-M Target Driver Setup

Debug Trace Flash Download Pack

Debug Adapter

Unit: ST-LINK/V2

☐ Shareable ST-Link

Serial

48FF6C067788494823362267

Version: HW: V2 FW: V2J38S7

☒ Check version on start

Target Com

Port: SW

Clock

Req: 10 MHz Selected: 1.800 MHz

Debug

Connect & Reset Options

Connect: Normal Reset: Autodetect

☒ Reset after Connect ☐ Stop after Reset

Cache Options

☒ Cache Code ☒ Cache Memory

Download Options

☐ Verify Code Download ☐ Download to Flash

SW Device

IDCODE	Device Name	Move
SWDIO	0x1BA01477 ARM CoreSight SW-DP (ARM Core	Up Down

☒ Automatic Detection ID CODE:

☐ Manual Configuration Device Name:

Add Delete Update IR len: AP: 0

确定 取消 应用(A)

打开指南者板上的
电源开关

Cortex-M Target Driver Setup

Debug | Trace | **Flash Download** | Pack

Download Function



- ☐ Erase Full Chip
- ☒ Erase Sectors
- ☐ Do not Erase
- ☒ Program
- ☒ Verify
- ☒ Reset and Run

RAM for Algorithm

Start: 0x20000000 Size: 0x00001000

Programming Algorithm

Description	Device Size	Device Type	Address Range
STM32F10x High-density ...	512k	On-chip Flash	08000000H - 0807FFFFH

Start: Size:

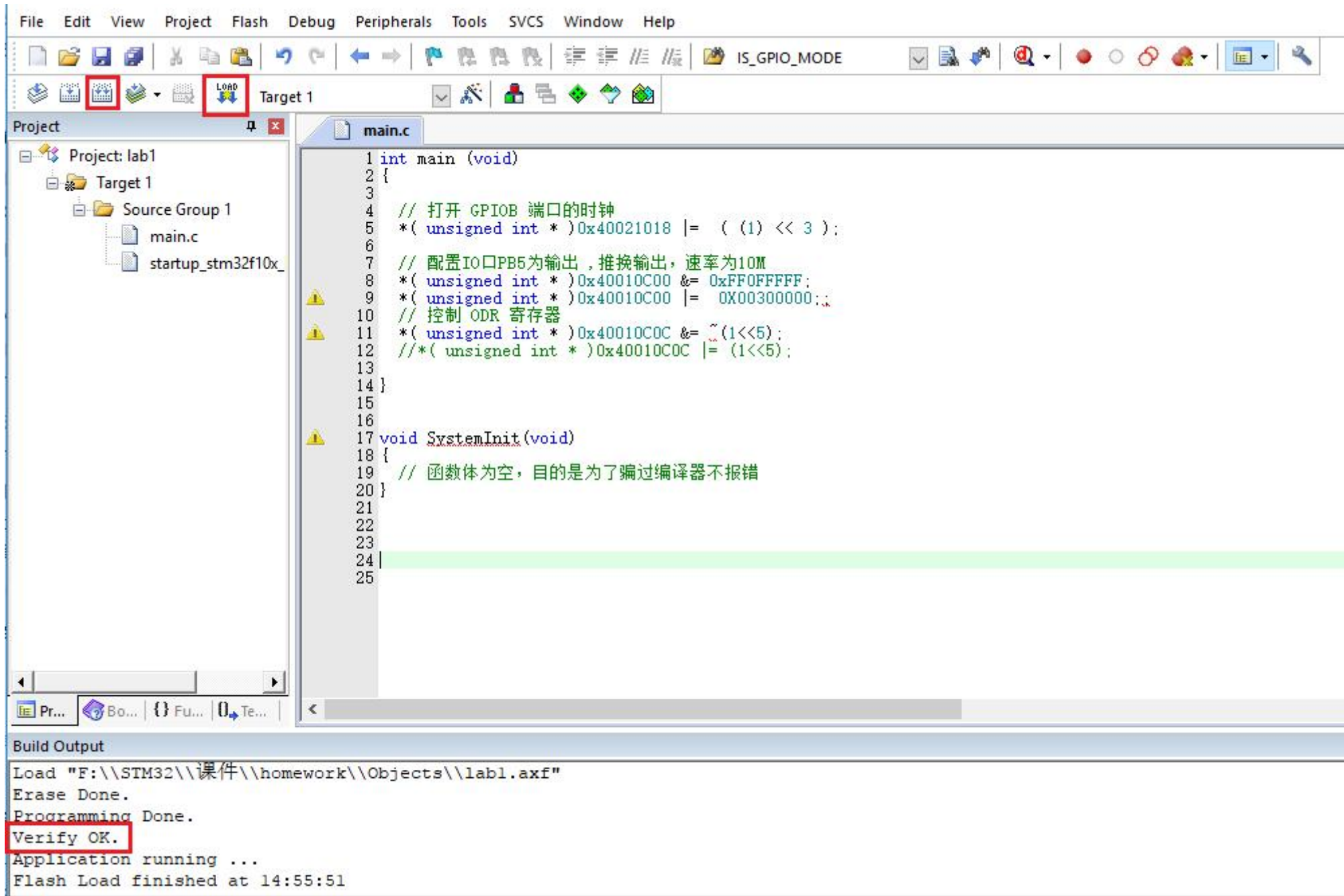
Add

Remove

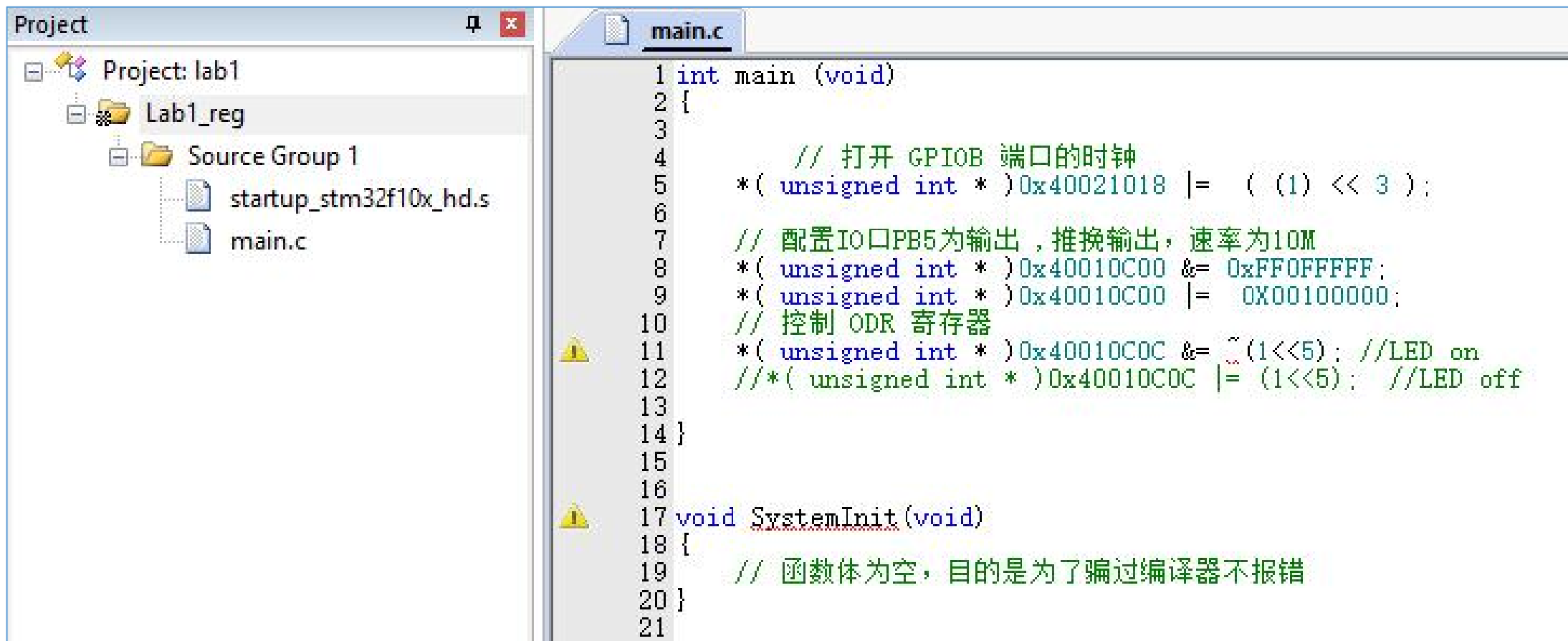
确定

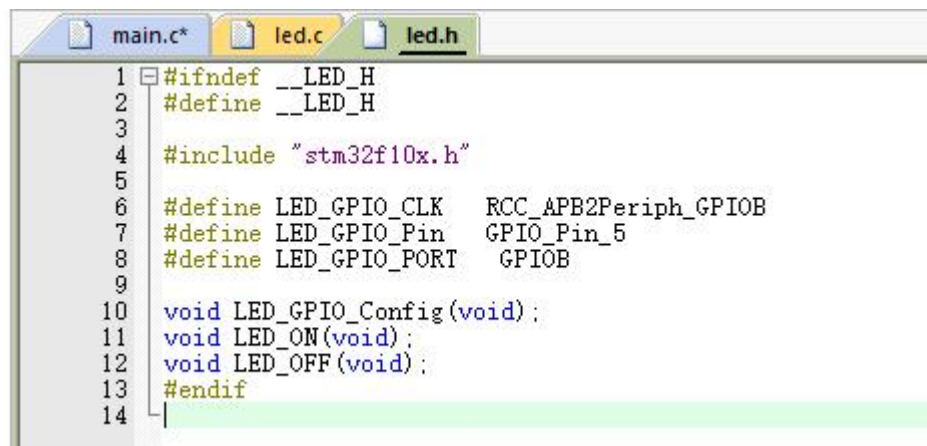
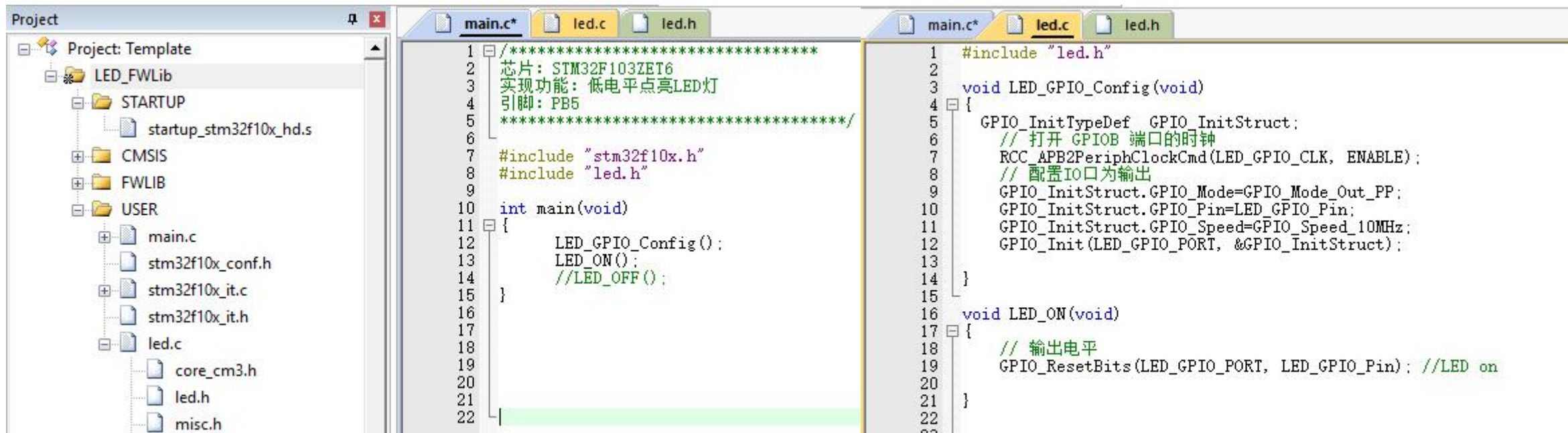
取消

应用(A)



Part V : Programming mode of STM32 & Code Analysis





Which one do you want to begin with ?

- ☐ A Lab1_reg1
- ☐ B LED_FWLib

提交

Programming mode of STM32

Register Programming: Process oriented

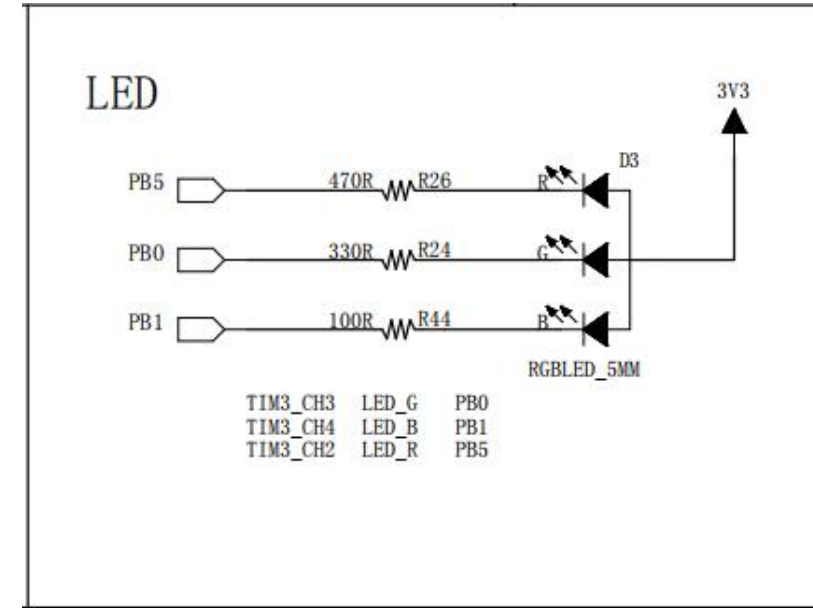


Firmware library programming: object oriented.
The library is to use C language to encapsulate the underlying operations, and users no longer need to care about register level operations.

Code Analysis

Design Purpose:

Light LED -> Corresponding IO port outputs low.



LED_Port $\xrightarrow[\text{diagram}]{\text{schematic}}$ PB5 $\left\{ \begin{array}{l} \text{outputs low, LED on;} \\ \text{outputs high, LED off;} \end{array} \right.$

LED_Port $\xrightarrow[\text{diagram}]{\text{schematic}}$ PB5 $\left\{ \begin{array}{l} \text{outputs low, LED on;} \\ \text{outputs high, LED off;} \end{array} \right.$

How ? ?

```
// 打开 GPIOB 端口的时钟
*( unsigned int * )0x40021018 |= ( (1) << 3 );

// 配置IO口PB5为输出,推挽输出, 速率为10M
*( unsigned int * )0x40010C00 &= 0xFF0FFFFFFF;
*( unsigned int * )0x40010C00 |= 0X00100000;
// 控制 ODR 寄存器
*( unsigned int * )0x40010C0C &= ~(1<<5); //LED on
/*( unsigned int * )0x40010C0C |= (1<<5); //LED off
```

Using any peripheral requires to enable its clock ! !
Clock enable must be before peripheral initialization

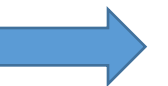
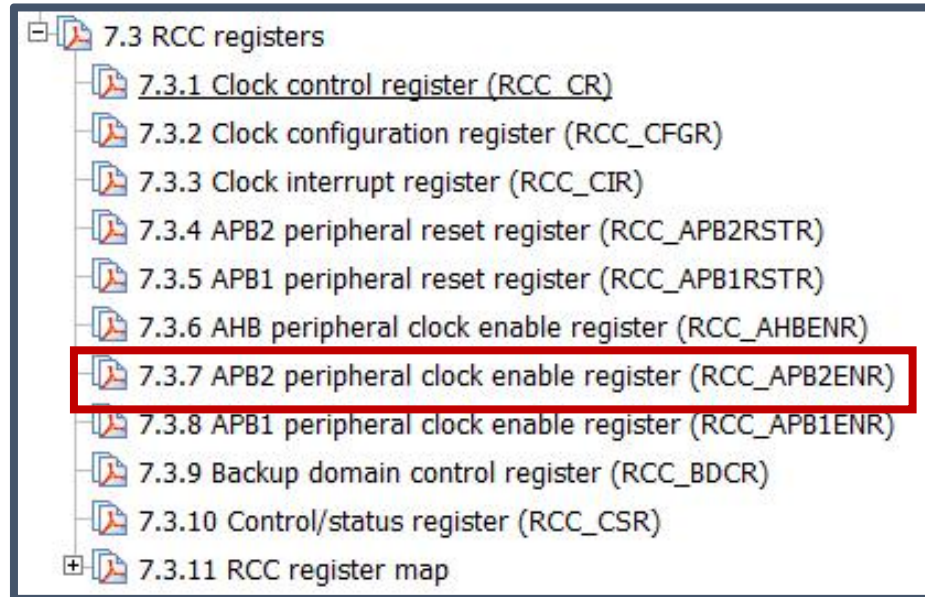


Table 3. Register boundary addresses

Boundary address	Peripheral	Bus	Register map
0xA000 0000 - 0xA000 0FFF	FSMC	AHB	Section 21.6.9 on page 564
0x5000 0000 - 0x5003 FFFF	USB OTG FS		Section 28.16.6 on page 913
0x4003 0000 - 0x4FFF FFFF	Reserved		-
0x4002 8000 - 0x4002 9FFF	Ethernet		Section 29.8.5 on page 1071
0x4002 3400 - 0x4002 7FFF	Reserved		-
0x4002 3000 - 0x4002 33FF	CRC		Section 4.4.4 on page 65
0x4002 2000 - 0x4002 23FF	Flash memory interface		-
0x4002 1400 - 0x4002 1FFF	Reserved		-
0x4002 1000 - 0x4002 13FF	Reset and clock control RCC		Section 7.3.11 on page 121
0x4002 0800 - 0x4002 0FFF	Reserved		-
0x4002 0400 - 0x4002 07FF	DMA2		Section 13.4.7 on page 289
0x4002 0000 - 0x4002 03FF	DMA1		
0x4001 8400 - 0x4001 FFFF	Reserved		-
0x4001 8000 - 0x4001 83FF	SDIO		Section 22.9.16 on page 621



```
// 打开 GPIOB 端口的时钟
*( unsigned int * )0x40021018 |= ( (1) << 3 );
```

7.3.7 APB2 peripheral clock enable register (RCC_APB2ENR)

Address: **0x18** **0x4002 1000 - 0x4002 13FF** **Reset and clock control RCC**

Reset value: 0x0000 0000

Access: word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TIM11 EN	TIM10 EN	TIM9 EN	Reserved		
										rw	rw	rw			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADC3 EN	USART 1EN	TIM8 EN	SPI1 EN	TIM1 EN	ADC2 EN	ADC1 EN	IOPG EN	IOPF EN	IOPE EN	IOPD EN	IOPC EN	IOPB EN	IOPA EN	Res.	AFIO EN
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw

1 Enable

Bit 3 **IOPBEN**: IO port B clock enable

Set and cleared by software.

0: IO port B clock disabled

1: IO port B clock enabled

Table 3. Register boundary addresses (continued)

Boundary address	Peripheral	Bus	Register map
0x4001 5800 - 0x4001 7FFF	Reserved		-
0x4001 5400 - 0x4001 57FF	TIM11 timer		Section 16.5.11 on page 468
0x4001 5000 - 0x4001 53FF	TIM10 timer		Section 16.5.11 on page 468
0x4001 4C00 - 0x4001 4FFF	TIM9 timer		Section 16.4.13 on page 458
0x4001 4000 - 0x4001 4BFF	Reserved		-
0x4001 3C00 - 0x4001 3FFF	ADC3		Section 11.12.15 on page 252
0x4001 3800 - 0x4001 3BFF	USART1		Section 27.6.8 on page 827
0x4001 3400 - 0x4001 37FF	TIM8 timer		Section 14.4.21 on page 363
0x4001 3000 - 0x4001 33FF	SPI1		Section 25.5 on page 742
0x4001 2C00 - 0x4001 2FFF	TIM1 timer		Section 14.4.21 on page 363
0x4001 2800 - 0x4001 2BFF	ADC2	APB2	Section 11.12.15 on page 252
0x4001 2400 - 0x4001 27FF	ADC1		Section 9.5 on page 194
0x4001 2000 - 0x4001 23FF	GPIO Port G		
0x4001 1C00 - 0x4001 1FFF	GPIO Port F		
0x4001 1800 - 0x4001 1BFF	GPIO Port E		
0x4001 1400 - 0x4001 17FF	GPIO Port D		
0x4001 1000 - 0x4001 13FF	GPIO Port C		
0x4001 0C00 - 0x4001 0FFF	GPIO Port B		
0x4001 0800 - 0x4001 0BFF	GPIO Port A		
0x4001 0400 - 0x4001 07FF	EXTI		Section 10.3.7 on page 214
0x4001 0000 - 0x4001 03FF	AFIO		Section 9.5 on page 194

9.2	GPIO registers
9.2.1	Port configuration register low (GPIOx_CRL) (x=A..G)
9.2.2	Port configuration register high (GPIOx_CRH) (x=A..G)
9.2.3	Port input data register (GPIOx_IDR) (x=A..G)
9.2.4	Port output data register (GPIOx_ODR) (x=A..G)
9.2.5	Port bit set/reset register (GPIOx_BSRR) (x=A..G)
9.2.6	Port bit reset register (GPIOx_BRR) (x=A..G)
9.2.7	Port configuration lock register (GPIOx_LCKR) (x=A..G)

```
// 配置IO口PB5为输出,推挽输出,速率为10M
*( unsigned int * )0x40010C00 &= 0xFF0FFFFF;
*( unsigned int * )0x40010C00 |= 0X00100000;
```

9.2.1 Port configuration register low (GPIOx_CRL) (x=A..G)

Address offset: 0x00

0x4001 0C00 - 0x4001 0FFF GPIO Port B

Reset value: 0x4444 4444

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF7[1:0]		MODE7[1:0]		CNF6[1:0]		MODE6[1:0]		CNF5[1:0]		MODE5[1:0]		CNF4[1:0]		MODE4[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	0001				rw	rw	rw	rw
15	14	13	12	11	10	9	8					3	2	1	0
CNF3[1:0]		MODE3[1:0]		CNF2[1:0]		MODE2[1:0]		CNF1[1:0]		MODE1[1:0]		CNF0[1:0]		MODE0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:30, 27:26, 23:22, 19:18, 15:14, 11:10, 7:6, 3:2 **CNFy[1:0]**: Port x configuration bits (y= 0 .. 7)
 These bits are written by software to configure the corresponding I/O port.
 Refer to [Table 20: Port bit configuration table](#).

In input mode (MODE[1:0]=00):

- 00: Analog mode
- 01: Floating input (reset state)
- 10: Input with pull-up / pull-down
- 11: Reserved

In output mode (MODE[1:0] > 00):

- 00: General purpose output push-pull
- 01: General purpose output Open-drain
- 10: Alternate function output Push-pull
- 11: Alternate function output Open-drain

Bits 29:28, 25:24, 21:20, 17:16, 13:12, 9:8, 5:4, 1:0 **MODEy[1:0]**: Port x mode bits (y= 0 .. 7)
 These bits are written by software to configure the corresponding I/O port.
 Refer to [Table 20: Port bit configuration table](#).

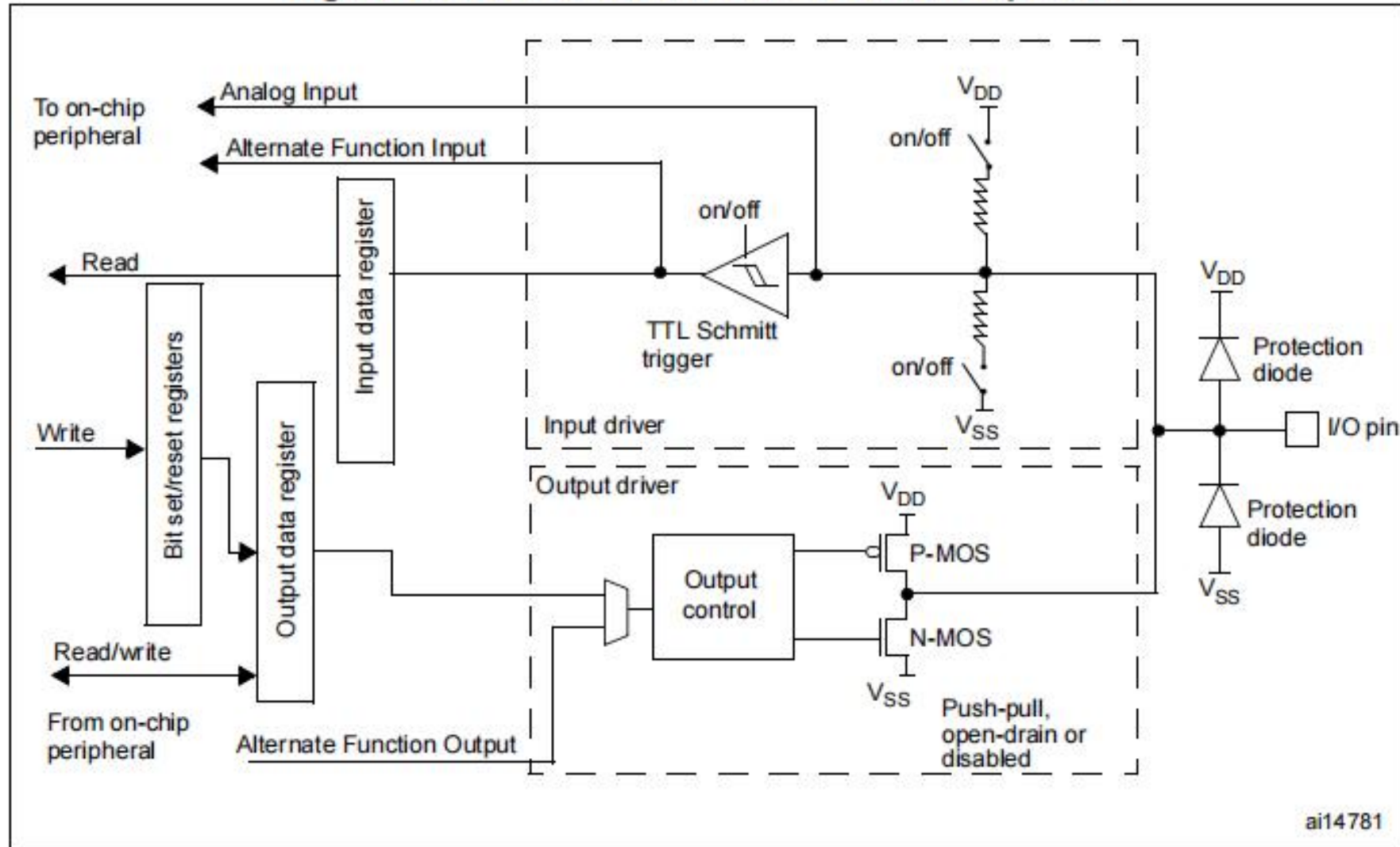
00: Input mode (reset state)

01: Output mode, max speed 10 MHz.

10: Output mode, max speed 2 MHz.

11: Output mode, max speed 50 MHz.

Figure 13. Basic structure of a standard I/O port bit



// 控制 ODR 寄存器

```
*( unsigned int * )0x40010C0C &= ~(1<<5); //LED on  
//*( unsigned int * )0x40010C0C |= (1<<5); //LED off
```

9.2.4 Port output data register (GPIOx_ODR) (x=A..G)

Address offset: **0x0C**

0x4001 0C00 - 0x4001 0FFF GPIO Port B

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0
r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w

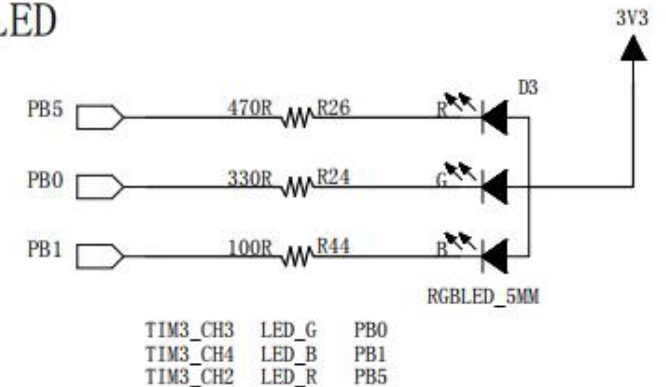
Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ODRy**: Port output data (y= 0 .. 15)

These bits can be read and written by software and can be accessed in Word mode only.

Note: For atomic bit set/reset, the ODR bits can be individually set and cleared by writing to the GPIOx_BSRR register (x = A .. G).

LED



0 LEDon

1 LEDoff

Register Name	Offset Address
CRL(32bit)	0x00
CRH(32bit)	0x04
IDR(32bit)	0x08
ODR(32bit)	0x0c
BSRR(32bit)	0x10
BRR(32bit)	0x14
LCKR(32bit)	0x18

0x4001 2000 - 0x4001 23FF	GPIO Port G
0x4001 1C00 - 0x4001 1FFF	GPIO Port F
0x4001 1800 - 0x4001 1BFF	GPIO Port E
0x4001 1400 - 0x4001 17FF	GPIO Port D
0x4001 1000 - 0x4001 13FF	GPIO Port C
0x4001 0C00 - 0x4001 0FFF	GPIO Port B
0x4001 0800 - 0x4001 0BFF	GPIO Port A

APB2

CRL: 0x4001 0C00+0x00
ODR:0x4001 0C00+0x0c

```
// 配置IO口PB5为输出，推挽输出，速率为10M
*( unsigned int * )0x40010C00 &= 0xFF0FFFFFF;
*( unsigned int * )0x40010C00 |= 0X00100000;
// 控制 ODR 寄存器
*( unsigned int * )0x40010C0C &= ~(1<<5); //LED on
*( unsigned int * )0x40010C0C |= (1<<5); //LED off
```

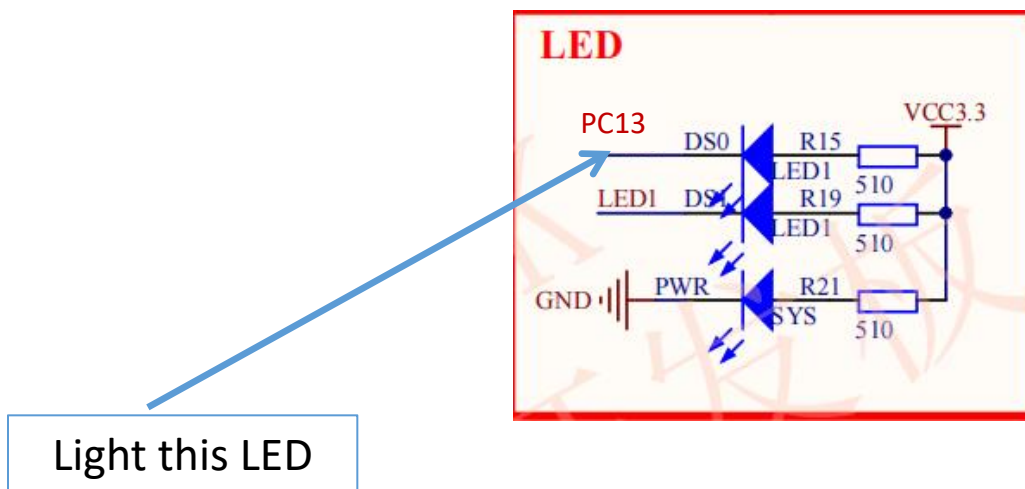
```
// 打开 GPIOB 端口的时钟
*( unsigned int * )0x40021018 |= ( (1) << 3 );

// 配置IO口PB5为输出 ,推挽输出, 速率为10M
*( unsigned int * )0x40010C00 &= 0xFF0FFFFFF;
*( unsigned int * )0x40010C00 |= 0X00100000;
// 控制 ODR 寄存器
*( unsigned int * )0x40010C0C &= ~(1<<5); //LED on
//*( unsigned int * )0x40010C0C |= (1<<5); //LED off
```

- 1, 先打开对应端口的时钟;
- 2, 配置输出, 确定输出模式; (配置时先将对应控制位清0, 后再根据需要进行配置)
- 3, 操作对应端口值

- 1：寄存器实质上就是_____，其赋值需要有_____。
- 2：STM32在使用外设前，必须在配置之前就打开其时钟。
STM32这么设计的原因是_____。
- 3：51单片机是否有该操作？

1：重新设计硬件，把LED接在PORTC-13口口，代码如何修改？



startup file

- The start-up file is necessary , and it is the entrance of the project;
- There is no need to modify, but to select the startup file matched with the chip used;
- Location: ..\官方固件库\STM32F10x_StdPeriph_Lib_V3.6.0 \Libraries \CMSIS\CM3 \DeviceSupport\ST\STM32F10x\startup\arm [Download URL](#)
- Which one : startup_stm32f10x_hd.s ➡

startup_stm32f10x_cl.s
startup_stm32f10x_hd.s
startup_stm32f10x_hd_vl.s
startup_stm32f10x_ld.s
startup_stm32f10x_ld_vl.s
startup_stm32f10x_md.s
startup_stm32f10x_md_vl.s
startup_stm32f10x_xl.s

启动文件	区别
startup_stm32f10x_ld.s	ld: low-density 小容量, FLASH 容量在 16-32K 之间
startup_stm32f10x_md.s	md: medium-density 中容量, FLASH 容量在 64-128K 之间
startup_stm32f10x_hd.s	hd: high-density 大容量, FLASH 容量在 256-512K 之间
startup_stm32f10x_xl.s	xl: 超大容量, FLASH 容量在 512-1024K 之间
以上四种都属于基本型, 包括 STM32F101xx、STM32F102xx、STM32F103xx 系列	
startup_stm32f10x_cl.s	cl:connectivity line devices 互联型, 特指 STM32F105xx 和 STM32F107xx 系列
startup_stm32f10x_ld_vl.s	vl:value line devices 超值型系列, 特指 STM32F100xx 系列
startup_stm32f10x_md_vl.s	
startup_stm32f10x_hd_vl.s	

Low-density devices are STM32F101xx, STM32F102xx and STM32F103xx microcontrollers where the Flash memory density ranges between 16 and 32 Kbytes.

Medium-density devices are STM32F101xx, STM32F102xx and STM32F103xx microcontrollers where the Flash memory density ranges between 64 and 128 Kbytes.

High-density devices are STM32F101xx and STM32F103xx microcontrollers where the Flash memory density ranges between 256 and 512 Kbytes.

XL-density devices are STM32F101xx and STM32F103xx microcontrollers where the Flash memory density ranges between 768 Kbytes and 1 Mbyte.

Connectivity line devices are STM32F105xx and STM32F107xx microcontrollers.

This section applies to low-, medium-, high- and XL-density STM32F10xxx devices. Connectivity line devices are dealt with in [Section 8: Connectivity line devices: reset and clock control \(RCC\)](#).



STM32 Standard Peripheral Libraries

STM32 Standard Peripheral Libraries Expansion

STM32 Standard Peripheral Drivers



F0



F1



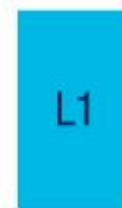
F3 F3



F2



F4



L1



■ STM32F37x/38x

■ STM32F301x/302x/303x/334x

<https://www.st.com/en/embedded-software/stm32-standard-peripheral-libraries.html>

<https://www.st.com/zh/embedded-software/stsw-stm32054.html>

startup_stm32f10x_hd.s

```

1  :***** (C) COPYRIGHT 2011 STMicroelectronics *****
2  : * File Name      : startup_stm32f10x_hd.s
3  : * Author        : MCD Application Team
4  : * Version       : V3.5.1
5  : * Date          : 08-September-2021
6  : * Description   : STM32F10x High Density Devices vector table for MDK-ARM
7  : *               : toolchain.
8  : *               : This module performs:
9  : *               : - Set the initial SP
10 : *               : - Set the initial PC == Reset_Handler
11 : *               : - Set the vector table entries with the exceptions ISR address
12 : *               : - Configure the clock system and also configure the external
13 : *               :   SRAM mounted on STM3210E-EVAL board to be used as data
14 : *               :   memory (optional, to be enabled by user)
15 : *               : - Branches to __main in the C library (which eventually
16 : *               :   calls main()).
17 : *               : After Reset the CortexM3 processor is in Thread mode,
18 : *               : priority is Privileged, and the Stack is set to Main.
19 : * <<< Use Configuration Wizard in Context Menu >>>
20 :*****

```

startup_stm32f10x_hd.s

```

147
148 ; Reset handler
149 Reset_Handler  PROC
150                 EXPORT Reset_Handler             [WEAK]
151                 IMPORT __main
152                 IMPORT SystemInit
153                 LDR     R0, =SystemInit
154                 BLX     R0
155                 LDR     R0, =__main
156                 BX      R0
157                 ENDP
158

```

Homework

- Install the softwares on your computer
- Realize the lighting experiment of LED
- Write a summary report:
 - ✓ The first lesson of introduction to STM32
 - ✓ complete the question in the class(modify LED to PC13)



Supplement 1—Pointer of C language

Pointer is the address;

*P----- A variable whose address is the content of P;

0x40010C00 is a value.

(unsigned int*) 0x40010C00, Cast type to pointer variable, that is
0x40010C00 is an address now ;

(unsigned int)0x40010C00 , (unsigned int*)0x40010C00 is p , so
(unsigned int)0x40010C00 , is the variable whos address is 0x40010C00 ;

